FLIPKART GRIDLOCK HACKATHON

Parking Intelligence

From reactive patrols to predictive enforcement.

Theme: Poor visibility on Parking-Induced congestion

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THE PROBLEM

On-street illegal parking is choking Bengaluru's traffic

- On-street illegal parking and spillover parking near commercial areas, metro stations, and events choke carriageways and intersections.
- One vehicle parked badly at a junction can back up traffic for blocks the same way one person blocking a doorway slows everyone behind them.
- This is the exact operational challenge defined in the hackathon brief not a generic “traffic is bad” framing.

FROM THE BRIEF

“On-street illegal parking and spillover parking near commercial areas, metro stations, and events choke carriageways and intersections.”

THE GAP

Why it's hard today

01

Patrol-based & reactive

Officers only learn about a hotspot by driving past it
or after a complaint there's no advance signal.

02

No violations-vs-impact heatmap

Nothing compares where violations happen against
how much they actually congest traffic.

03

Can't prioritize zones

With limited patrol units, there's no ranked answer to
“which hotspot deserves attention first.”

OUR SOLUTION

248,000 tickets → 1,323 ranked, patrol-sized hotspots

Ranked not by how many violations happened but by how much each hotspot actually disrupts traffic flow.

Raw violations
(248,691 cleaned rows)

→

Spatial clustering
(DBSCAN → 1,323 zones)

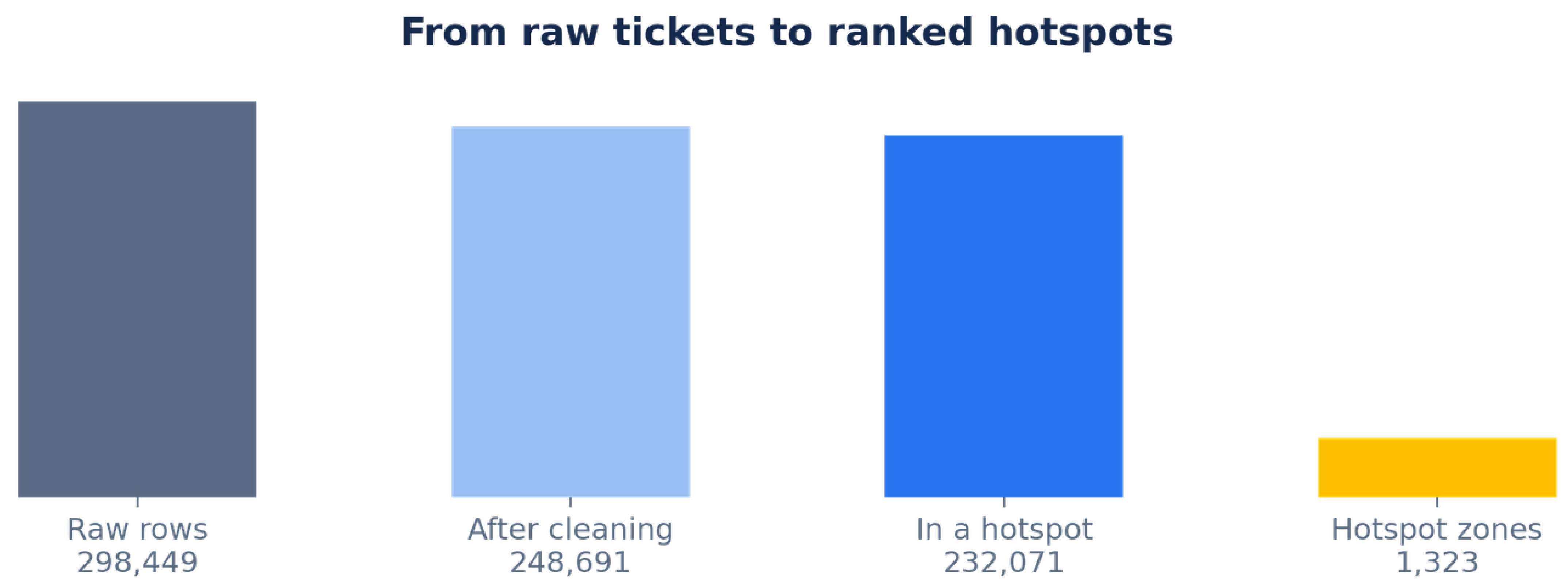
→

Impact scoring
(Hotspot × Congestion × Priority)

Result: a live, filterable dashboard an officer can use to decide where to send the next patrol in minutes, not guesswork.

THE PIPELINE

How it works



1. Clean & cluster

DBSCAN groups nearby violations into patrol-sized zones adapts to real street shapes, no rigid grid.

2. Score

Every zone gets a Hotspot Score, a Congestion Impact score, and a blended Priority Score.

3. Serve & visualize

A live dashboard, filterable by time, jurisdiction, and vehicle/violation type.

From points to patrol-sized zones

WHY NOT A SIMPLE GRID?

- A fixed grid cell cuts a real hotspot in half at the boundary undercounting it in both cells.
- Wastes cells on empty space and can't adapt to real, non-square street shapes.

OUR CHOICE: DBSCAN

- Stand at any violation: ≥ 12 others within 150 m means a dense, real cluster otherwise it's noise.
- Finds any-shaped clusters, ignores one-offs, and needs no preset cluster count (unlike k-means).

THE FIX WE ADDED — RECURSIVE PATROL-SIZING

Plain DBSCAN chains dense corridors into one 3 km mega-cluster no single patrol can cover.
So whenever a cluster's radius exceeds 200 m, we re-run DBSCAN on just those points with a smaller search radius recursively until every zone is patrol-sized. → 1,323 patrol-sized zones.

Three scores, recomputed live per filter

HOTSPOT SCORE

Is this a real, repeated problem?

log(volume) blended with violation density per unit area the log keeps one mega-zone from flattening every other zone to zero.

CONGESTION IMPACT

How much does it choke traffic?

A weighted blend of three signals:

- Junction proximity — 45%
- Obstruction severity (per type) — 40%
- Peak-hour concentration — 15%

PRIORITY SCORE

Where do we send the patrol?

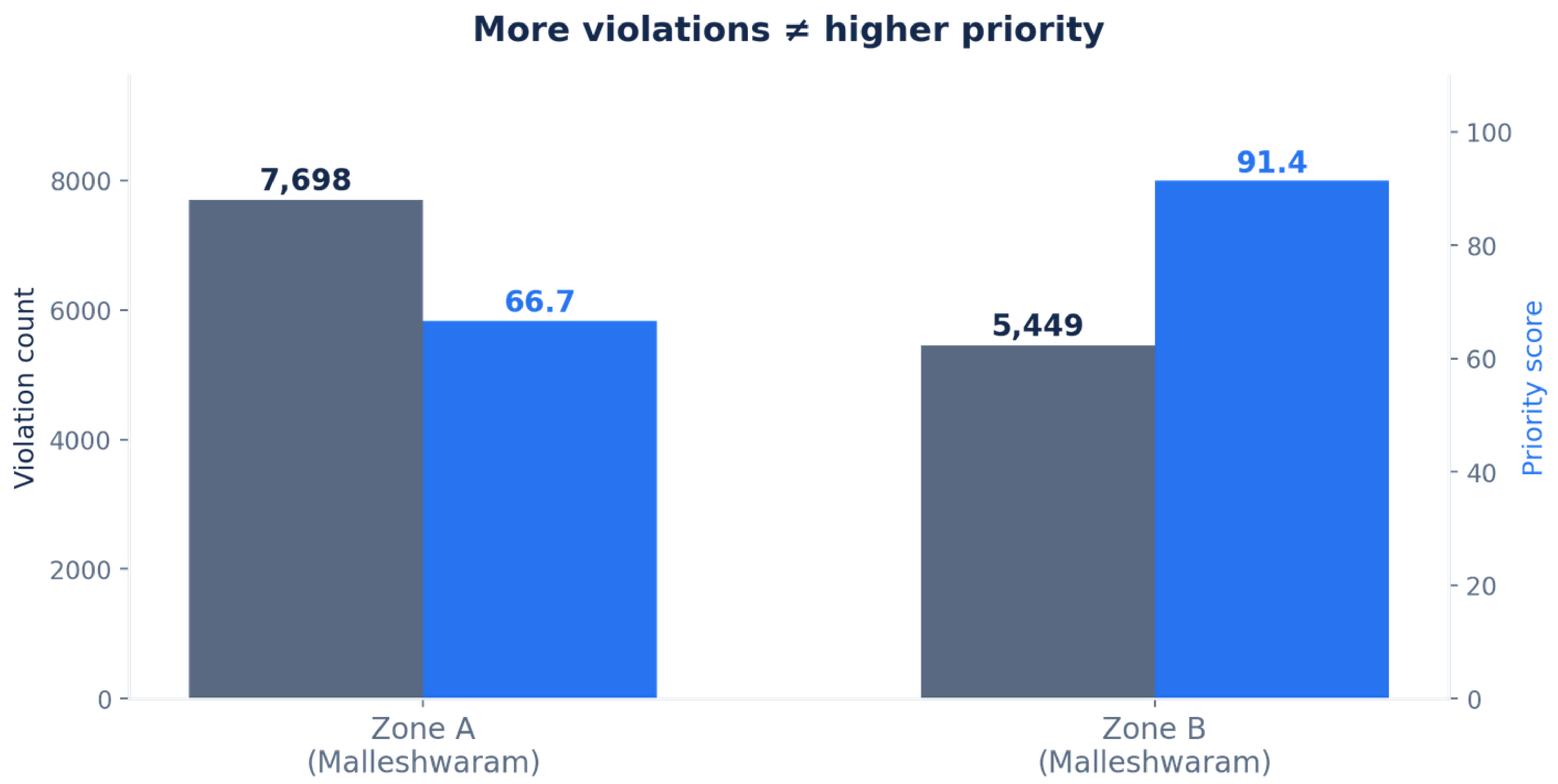
0.45 × Hotspot + 0.55 × Congestion. Deliberately favours impact over raw volume the exact ask in the brief.

OBSTRUCTION SEVERITY — A HAND-BUILT TABLE PER VIOLATION TYPE (0 = no flow impact, 1 = severe)

Main-road / double-parking = 1.0 · near road-crossing = 0.95 · wrong parking = 0.70 · no parking = 0.55 · on footpath = 0.40 · defective number plate = 0.0

THE CORE IDEA

Volume ≠ Impact



WHY THIS WINS

- Zone A has 41% more violations than Zone B.
- But Zone B sits on a busier junction at peak hours so it outranks Zone A on priority.
- This is the exact distinction the brief asks for: violations vs. congestion impact.

SEE IT IN ACTION

Live demo

DEMO LINK: https://bengaluru-parking-intelligence-beryl.vercel.app/

BY THE NUMBERS

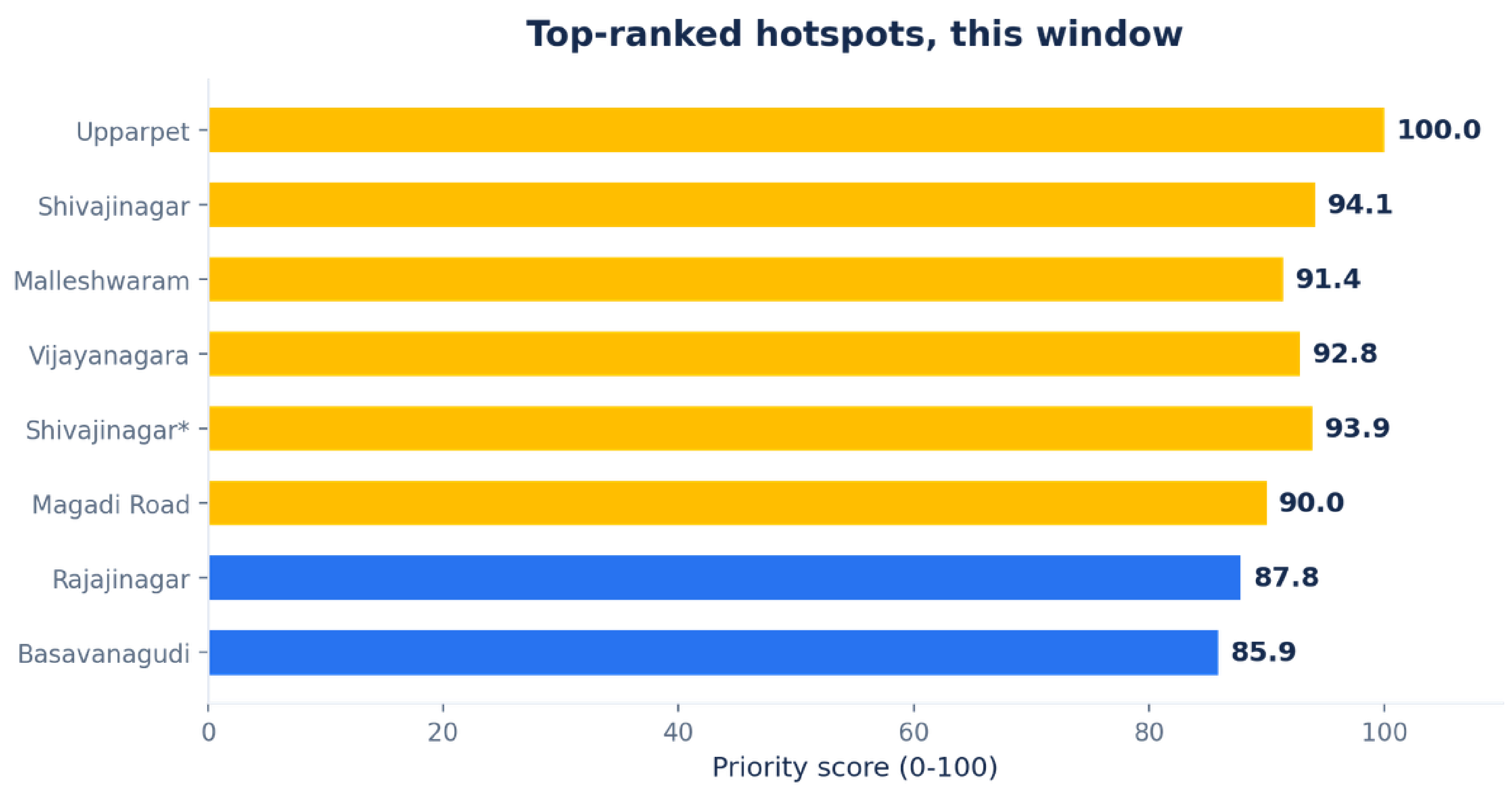
Results

1,323 hotspot zones found

85 flagged high-priority right now

42,707 violations in the #1 zone alone

93.3% of all violations mapped to a real hotspot



What's next

Validate with real traffic data

Fit congestion weights against actual traffic-flow data (signal timing, GPS speed) once available.

Patrol routing

Turn the ranked list into an actual route for available patrol units — vehicle routing on top of prioritization.

Feedback loop

Let a patrol mark a hotspot “cleared” and feed that back to suppress its ranking going forward.

From reactive patrols to predictive enforcement.

248,000 tickets → 1,323 ranked hotspots → one defensible answer to ‘where next.’

Team Inferix · Thank you